

Hypothesis Testing in the Linear Regression Model

Lecture 4

This lecture introduces classical tests for the linear regression model. To be consistent with the previous lectures, we assume that the model is:

$$y = X\beta + u, \quad (1)$$

where the response variable y is an n -dimensional vector, X is a $n \times p$ matrix, β is a p dimensional vector and u is a n dimensional error term. We are interested in inference on the parameter of interest, β .

Recall the following assumptions:

- A1 The model is linear and given by equation (1).
- A2 The rank of the matrix X is p .
- A3 The conditional expected value of y is equal to $E(y_i|X) = x_i'\beta$ for all subjects i in the sample of size n .
- A4 The conditional variance of y is equal to $V(y|X) = V(X\beta + u|X) = V(u) = \sigma^2 I$, where $\sigma^2 > 0$ and I is the identity matrix.

Assumption A1 implies that the model is correctly specified and A2 implies that the projection matrix $P = X(X'X)^{-1}X'$ exists. Note that this also implies that the OLS estimator, $\hat{\beta} = (X'X)^{-1}X'y$ exists too. A3 implies that the error term u and the independent variables are not correlated:

$$E(u_i|X) = E(y_i - x_i'\beta|X) = E(y_i|X) - x_i'\beta = x_i'\beta - x_i'\beta = 0$$

Note that this also implies that $E(u_i) = E(E(u_i|X)) = 0$ and the covariance between independent variables and error term is zero. Lastly, A4 implies that the error term u has constant variance, σ^2 , and u_i and u_j are not correlated.

We add one more assumption that will be relaxed in Lecture 5:

A5 The conditional distribution of y is normal with mean $X\beta$ and variance $\sigma^2 I$.

In this lecture, we will discuss how to evaluate different hypothesis on the coefficients in the following regression model,

$$y_i = x_i' \beta + u_i = \beta_0 + \sum_{j=1}^p x_{ij} \beta_j + u_i.$$

For instance, the hypotheses we might want to evaluate are,

$$\begin{aligned} H_0 : \quad & \beta_1 = 0 \\ H_0 : \quad & \beta_1 = \beta_2 \\ H_0 : \quad & \beta_p + \beta_{p-1} = c \\ H_0 : \quad & \beta_1 = \beta_2 = \dots = \beta_p = 0. \end{aligned}$$

1 Linear hypothesis testing

A general expression for linear hypothesis is of the following form:

$$\begin{aligned} H_0 : R\beta &= \delta, \\ H_a : R\beta &\neq \delta \end{aligned}$$

where H_0 is the null hypothesis, H_a is the alternative hypothesis, R is a known $r \times p$ matrix with rank r , and δ is a $r \times 1$ vector. Here r indicates the number of restrictions applied to p parameters.

Example 1. Suppose $\beta = (\beta_1, \beta_2, \beta_3)$ in equation (1). Therefore,

$$\begin{aligned} r_{11}\beta_1 + r_{12}\beta_2 + r_{13}\beta_3 &= \delta_1, \\ r_{21}\beta_1 + r_{22}\beta_2 + r_{23}\beta_3 &= \delta_2. \end{aligned}$$

Suppose we want to test $\beta_1 + \beta_2 = 1$ and $\beta_1 - \beta_3 = 0$. Then,

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}; \quad \delta = \begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Note that the rank of R is 2.

Example 2. The simplest conceivable case is $r = 1$, then $R = [1, 1, 0]$, $\delta = 1$, and then test

$$H_0 : \beta_1 + \beta_2 = 1$$

How we can test the previous null hypothesis? Under Assumption A5, we have that

$$\begin{aligned}\hat{\beta} &\sim \mathcal{N}(\beta, \sigma^2(X'X)^{-1}) \\ \hat{\beta} - \beta &\sim \mathcal{N}(0, \sigma^2(X'X)^{-1}) \\ \sigma(X'X)^{1/2}\hat{\beta} - \beta &\sim \mathcal{N}(0, I_p).\end{aligned}$$

It can be shown then that,

$$R\hat{\beta} \sim N(R\beta, \sigma^2 R(X'X)^{-1}R').$$

Therefore, in the case that $r = 1$,

$$\frac{R\hat{\beta} - R\beta}{\sqrt{\sigma^2 R(X'X)^{-1}R'}} \sim N(0, 1).$$

Under the null hypothesis, it is equal to

$$\frac{R\hat{\beta} - \delta}{\sqrt{\sigma^2 R(X'X)^{-1}R'}} \sim N(0, 1)$$

Note however that this test statistic is unfeasible because σ^2 is unknown. We replace σ^2 by $\hat{\sigma}^2$ defined as,

$$\hat{\sigma}^2 = \frac{\hat{u}'\hat{u}}{n - p}$$

Why? Note that the estimator $\hat{\sigma}^2$ is an unbiased and consistent estimator for σ^2 . Let's write first the matrix $P = X(X'X)^{-1}X'$ and $M = I - P$. M is symmetric and idempotent

$$MM = M^2 = M$$

and $MX = (I - X(X'X)^{-1}X')X = X - X = 0$. Note that

$$\begin{aligned}E(\hat{u}'\hat{u}) &= E((y - X\hat{\beta})'(y - X\hat{\beta})) = E((y - Py)'(y - Py)) \\ &= E((My)'(My)) = E((Mu)'(Mu)) = E(u'Mu) \\ &= E(\text{trace}(u'Mu)) = \text{trace}(E(u'u)M) = \text{trace}(E(uu')M) \\ &= \text{trace}(\sigma^2 IM) = \sigma^2 \text{trace}(M) \\ &= \sigma^2 \text{trace}(I - X(X'X)^{-1}X') = \sigma^2(n - p).\end{aligned}$$

Therefore, $E\hat{u}'\hat{u}/n - p = \sigma^2$.

It is also consistent,

$$\hat{\sigma}^2 = \frac{u'u}{n-p} - \frac{u'X(X'X)^{-1}X'u}{n-p}$$

The first term is, under A4,

$$\frac{\sum_{i=1}^n u_i^2}{n-p} \rightarrow \sigma^2.$$

The second term, under A3,

$$\frac{1}{n-p} \sum_{i=1}^n x_i' u_i \rightarrow 0$$

We know that $\frac{1}{n}(X'X) \rightarrow Q_0$, and then by continuity of inverse operation, we have that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n}(X'X) \right)^{-1} \rightarrow Q_0^{-1}.$$

By Slutsky, we have that,

$$\begin{aligned} \lim_{n \rightarrow \infty} \hat{\sigma}^2 &= \lim_{n \rightarrow \infty} \left(\frac{u'u}{n-p} - \frac{u'X(X'X)^{-1}X'u}{n-p} \right) \\ &= \sigma^2 \end{aligned}$$

The distribution of our test statistic is t -student with $n-r$ degrees of freedom. Note that,

$$\begin{aligned} \frac{R\hat{\beta} - \delta}{\sqrt{\hat{\sigma}^2 R(X'X)^{-1}R'}} &= (\hat{\sigma}^2 R(X'X)^{-1}R')^{-1/2} (R\hat{\beta} - \delta) \\ &= ((\hat{u}'\hat{u}/(n-p))R(X'X)^{-1}R')^{-1/2} (R\hat{\beta} - \delta) \\ &= ((\hat{u}'\hat{u}/(n-p)\sigma^2)\sigma^2 R(X'X)^{-1}R')^{-1/2} (R\hat{\beta} - \delta) \\ &= (\hat{u}'\hat{u}/(n-p)\sigma^2)^{-1/2} (\sigma^2 R(X'X)^{-1}R')^{-1/2} (R\hat{\beta} - \delta) \\ &= (\hat{u}'\hat{u}/(n-p)\sigma^2)^{-1/2} \times \mathcal{N}(0, 1) \\ &= \frac{\mathcal{N}(0, 1)}{\sqrt{\frac{(n-r)\hat{\sigma}^2}{\sigma^2}/n-r}} \\ &= \frac{N(0, 1)}{\sqrt{\chi_{n-r}^2/n-r}}, \end{aligned}$$

because $\hat{\sigma}^2$ and $\hat{\beta}$ are independent.

Example 3. Suppose $R = (1, \dots, 0)$ and $\delta = 0$.

$$R\hat{\beta} - \delta = \hat{\beta}_1$$

and $R(X'X)^{-1}R' = (X'X)_{ii}^{-1}$. Therefore, we can test $H_0 : \beta_1 = 0$, by,

$$\frac{\hat{\beta}_1}{\hat{\sigma}\sqrt{(X'X)_{ii}}} = \frac{\hat{\beta}}{se(\hat{\beta}_1)} \sim t_{n-1}.$$

where se denotes standard error. The null hypothesis $H_0\beta_j = 0$ can be tested by considering

$$\frac{\hat{\beta}_j}{se(\hat{\beta}_j)}.$$

Example 4. Consider the model,

$$wages_i = \beta_0 + \beta_1 educ_i + \beta_2 exp_i + \beta_3 tenure_i + u_i$$

Consider the example in Wooldridge's book.

$$\begin{array}{rcccc} \log(wage) & = & 0.284 & + & 0.092educ & + & 0.0041exp & + & 0.022tenure \\ & & (0.104) & & (0.007) & & (0.0017) & & (0.003) \end{array}$$

We would like to test if experience is a significant factor to explain wages. Consider:

$$H_0 : \beta_{exp} = 0; \quad H_a : \beta_{exp} > 0$$

Our t statistic is

$$\hat{t} = \frac{\hat{\beta}_{exp}}{se(\hat{\beta}_{exp})} = \frac{0.0041}{0.0017} = 2.4$$

Since the degrees of freedom is $n - p - 1 = 526 - 3 - 1 = 522$, we can use the standard normal critical values. Then $\hat{t} = 2.4 > 1.64 = c$ we reject the null hypothesis, and we conclude that experience is statistically significantly different from zero at 5 percent. We conclude also that $2\frac{1}{2}$ years of experience would increase wages by 1 percent.

Alternatively we consider the null and alternative hypotheses,

$$H_0 : \beta = 0; \quad H_a : \beta \neq 0$$

In this case, economic theory does not help to determine if the parameter should be positive or negative, so the alternative hypothesis reflects the lack of information. The rejection rule is,

$$|t| > c$$

Note that we are considering the absolute value of the t statistic. If we reject the null hypothesis, we say that x is statistically significantly different from zero. If we cannot reject the null hypothesis that $\beta = 0$, we say that x is not statistically significantly different from zero. We never say that we accept the null hypothesis.

INSERT FIGURE ONE and TWO-SIDED TEST

Example 5. Suppose you derive that the demand equation of a good is $Q = c/P$, where Q is quantity, P is price, and c is a positive constant. In this case, the demand elasticity is -1. We can formulate a regression version of the demand to test

$$\log(Q) = \log(c) + \beta \log(P) + u$$

The null hypothesis is $H_0 : \beta = -1$. Under the null hypothesis, the appropriate t statistic is

$$\hat{t} = \frac{\hat{\beta} + 1}{se(\hat{\beta})}$$

Remark 1. (P-values) It is defined as the smallest significance level at which the null hypothesis would be rejected. For instance, if the t statistic is $\hat{t} = 0.0039/0.0022 = 1.77$, we can compute the p-value as

$$\text{p-value} = P(T > 1.77) = 1 - 0.9616 = 0.0384$$

Therefore, we conclude that we reject the null hypothesis at 10 and 5 percent, but we cannot reject the null hypothesis at 1 percent significance level.

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What if we have r independent restrictions? Consider first the following two results:

If $Z \sim N(M, \Omega)$ and $k = \text{rank}(\Omega)$, then

$$(Z - M)' \Omega^{-1} (Z - M) \sim \chi_k^2.$$

If χ_1^2 , has n_1 degrees of freedom and χ_2^2 has n_2 degrees of freedom, then,

$$\frac{\chi_1^2/n_1}{\chi_2^2/n_2} \sim F_{n_1, n_2}.$$

By result 1, because $R\hat{\beta} \sim N(R\beta, \sigma^2 R(X'X)^{-1}R')$, we have that,

$$(R\hat{\beta} - \delta)' \sigma^{-2} (R(X'X)^{-1}R')^{-1} (R\hat{\beta} - \delta) \sim \chi_r^2$$

The feasible version of the test statistic includes $\hat{\sigma}^2$, and therefore, it can be written as,

$$\begin{aligned} (R\hat{\beta} - \delta)' \frac{1}{\hat{\sigma}^2} [R(X'X)^{-1}R']^{-1} (R\hat{\beta} - \delta) / r &= \frac{(R\hat{\beta} - \delta)' \frac{1}{\hat{\sigma}^2} (R(X'X)^{-1}R')^{-1} (R\hat{\beta} - \delta) / r}{(u'Mu/\sigma^2)/n - p} \\ &= \frac{(R\hat{\beta} - \delta)' (R(X'X)^{-1}R')^{-1} (R\hat{\beta} - \delta)}{r\hat{\sigma}^2} \sim F_{r, n-p}. \end{aligned}$$

The test statistic is,

$$F = \frac{(R\hat{\beta} - \delta)' (R(X'X)^{-1}R')^{-1} (R\hat{\beta} - \delta)}{r\hat{\sigma}^2} \quad (2)$$

Example 6. Consider that we want to impose q restrictions e.g. $\beta_1 = \beta_2 = \dots = \beta_q = 0$. In other words, we want to evaluate if the group of variables $x_1, x_2, \dots, x_q$ has no significant effect on the dependent variable. Our null hypothesis will be

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_q = 0 \quad H_a : \text{at least one } \beta_j \neq 0,$$

which suggests that under the null hypothesis the true model is

$$y = \beta_0 + \beta_{q+1}x_{q+1} + \dots + \beta_px_p + u \quad (3)$$

instead of the model

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_qx_q + \dots + \beta_px_p + u \quad (4)$$

The first model is the restricted model because we are imposing q restrictions. The second model is the unrestricted since we are not imposing a restriction. To evaluate the null hypothesis, we use the F test, which is defined as

$$F \equiv \frac{(SSR_r - SSR_u)/q}{SSR_u/(n - p - 1)}$$

where SSR_r is the sum of squared residuals from the restricted model and SSR_u is the sum of squared residuals from the unrestricted model. The F statistic and the R^2 of a regression has a close relation since both statistics depend on the sum of squared residuals. The derivation of the exact relation is left as an exercise.

Example 7. It is useful to evaluate the group of variables $x_1, x_2, \dots, x_q$ has no effect on the

dependent variable, which is basically an evaluation of the regression model. This is the case where none of the explanatory variables has a significant effect on the dependent variable,

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \quad H_a : \text{at least one } \beta_j \neq 0,$$

When we reject the null hypothesis using an F test, we say that the group of variables are jointly statistically significant at the appropriate significance level.

2 Performance of tests under non-Gaussian assumptions

The classical t and F statistic might be used under non-normality conditions. However, as it will be clear in the cases below, a necessary condition is that the sample should be reasonable large. Another important change is associated with the critical values. Instead of looking at the t distribution and F distribution for critical values, we consider the standard normal distribution for the t test and the chi-square distribution for the F test.

Assume that we are interested in the j -th covariate in a multivariate regression. We are interested in the hypothesis $\beta_j = 0$ is

$$t = \frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 (X'X)^{-1}_{jj}}} = \frac{\sqrt{n}\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 n (X'X)^{-1}_{jj}}}$$

where $(X'X)^{-1}_{jj}$ denotes the (j, j) -th element of the matrix $X'X^{-1}$. Under the null hypothesis that $\beta_{j0} = 0$, we have,

$$\sqrt{n}(\hat{\beta}_j) \rightarrow \mathcal{N}(0, \sigma_0^2 Q_{0,jj}^{-1}),$$

where Q_0 is the limiting form of the $X'X$. Then, because $\hat{\sigma}^2$ is a consistent estimator for the parameter σ_0^2 and $(X'X)/n$ is a consistent estimator for the unknown matrix Q_0 , we can apply the Slutsky theorem and obtain:

$$t = \frac{\sqrt{n}\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 n (X'X)^{-1}_{jj}}} \rightarrow \mathcal{N}(0, 1)$$

The argument for the F test is similar. Under the null hypothesis $R\beta_0 = r$, we have that

$$\sqrt{n}(R\hat{\beta} - r) = \sqrt{n}(R\hat{\beta} - R\beta_0) = \sqrt{n}R(\hat{\beta} - \beta_0) \rightarrow N(0, \sigma_0^2(RQ_0^{-1}R'))$$

Note also that $\hat{\sigma}^2 \rightarrow \sigma_0^2$ and

$$\frac{1}{n}[R(X'X)^{-1}R']^{-1} \rightarrow (RQ_0^{-1}R')^{-1}$$

Therefore

$$\hat{F} = \frac{(R\hat{\beta} - r)'[R(X'X)^{-1}R']^{-1}(R\hat{\beta} - r)/(p - k)}{\hat{\sigma}^2} \rightarrow \chi_{p-k}^2$$